

Himanshu Mishra

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Education

Babu Banarasi Das Institute of Technology and Management (BBDITM), Lucknow *Apr 2024 – Apr 2028*
Bachelor of Technology in Computer Science (Penultimate Year Candidate)

- Relevant Coursework: Data Structures, Algorithms, Distributed Systems, Operating Systems, Computer Networks, Database Management
- Active member of coding club participating in hackathons, algorithm analysis, and systems workshops

Technical Skills

Programming Languages: C++, Java, Python, Typescript, JavaScript, SQL, HTML/CSS

Core Infrastructure: Distributed Systems, Concurrency Control, Multi-threading, Synchronization, Sockets (TCP/IP)

Systems & Databases: SQLite WAL, MongoDB, PostgreSQL, RESTful APIs, WebSockets, Node.js, Express, React, Next.js

Developer Tools & Practices: Git, GitHub, VS Code, Postman, Docker, NPM, CI/CD, Agile/Scrum

Projects

HyperQueue: Distributed Task Queue Node.js, SQLite, TCP, WebSockets [github](#) 

- Designed and built a **high-performance**, multi-threaded background task manager using raw TCP sockets and Node.js `worker_threads` processing up to **10K tasks/min** without blocking event loops.
- Implemented robust SQLite state persistence using Write-Ahead Logging (WAL) and transactions for **fault-tolerant** crash-recovery.
- Programmed custom queue **synchronization** algorithms featuring delay scheduling, priorities, and exponential backoff retry models.
- Created a real-time monitor console via WebSockets to visualize queue performance metrics, worker loads, and system debugging logs.

HyperDocs: Collaborative Document Editor Node.js, WebSockets, SQLite, Gemini API [github](#) 

- Engineered a real-time collaborative workspace leveraging Operational Transformations (OT) to resolve concurrent insertion/deletion conflicts, **reducing conflict resolution latency by 40%** and guaranteeing state convergence.
- Built WebSockets infrastructure to sync selection highlights and cursor coordinates in real-time, supporting up to **50 concurrent editors**.
- Integrated a **high-performance** Gemini AI co-pilot helper using secure API proxies for automated content refactoring, grammar analysis, and autocompletion inline.
- Logged document histories and version logs in an SQLite database to ensure reliable document backups and state restoration.

CRM Application [Live](#)  [github](#) 

- Developed a customer relationship management system featuring lead tracking pipelines, multi-user role-based access, and JWT session handling.
- Optimized database search latency by implementing smart indexing on MongoDB, lowering lead query response time to sub-100ms for over **1,000 lead records**.
- Implemented dashboard analytics and live pipelines to monitor user interactions, delegate tasks, and audit logs.

Experience

Freelance Full Stack Developer Self-Employed (Remote) *Oct 2025 – Present*

- Delivered 5+ client applications including portfolio websites, business dashboards, and custom backend systems.
- Collaborated with clients to analyze requirements, evaluate design alternatives, choose technical stacks, and resolve system issues.

Achievements & Certifications

- **Problem Solving:** Solved 300+ DSA problems across LeetCode, CodeChef, and GeeksforGeeks platforms, focusing on trees, graphs, and dynamic programming.
- Strong foundation in core computer science subjects including Operating Systems, OOPS, Database Management, and Networking.
- **Open Source:** Contributor to open-source developer tools, troubleshooting issues, and submitting verified pull requests.
- **Certifications:** Full Stack Web Development (Udemy), Google hack2skills.